



The Coin Change Problem

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Problem

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Given an amount and the denominations of coins available, determine how many ways change can be made for amount. There is a limitless supply of each coin type.

Example

$n = 3$

$c = [8, 3, 1, 2]$

There are **3** ways to make change for $n = 3$: $\{1, 1, 1\}$, $\{1, 2\}$, and $\{3\}$.

Function Description

Complete the `getWays` function in the editor below.

`getWays` has the following parameter(s):

- `int n`: the amount to make change for
- `int c[m]`: the available coin denominations

Returns

- `int`: the number of ways to make change

Input Format

The first line contains two space-separated integers n and m , where:

n is the amount to change

m is the number of coin types

The second line contains m space-separated integers that describe the values of each coin type.

Constraints

- $1 \leq c[i] \leq 50$
- $1 \leq n \leq 250$
- $1 \leq m \leq 50$
- Each $c[i]$ is guaranteed to be distinct.

Hints

Solve overlapping subproblems using [Dynamic Programming \(DP\)](#):

You can solve this problem recursively but will not pass all the test cases without optimizing to eliminate the [overlapping subproblems](#). Think of a way to

store and reference previously computed solutions to avoid solving the same subproblem multiple times. * Consider the degenerate cases:

- How many ways can you make change for **0** cents? - How many ways can you make change for **> 0** cents if you have no coins? * If you're having trouble defining your solutions store, then think about it in terms of the base case (**$n = 0$**). - The answer may be larger than a **32**-bit integer.

Sample Input 0

```
4 3
1 2 3
```

Sample Output 0

```
4
```

Explanation 0

There are four ways to make change for **$n = 4$** using coins with values given by **$C = [1, 2, 3]$** :

1. **$\{1, 1, 1, 1\}$**
2. **$\{1, 1, 2\}$**
3. **$\{2, 2\}$**
4. **$\{1, 3\}$**

Sample Input 1

```
10 4
2 5 3 6
```

Sample Output 1

```
5
```

Explanation 1

There are five ways to make change for **$n = 10$** units using coins with values given by **$C = [2, 5, 3, 6]$** :

1. **$\{2, 2, 2, 2, 2\}$**
2. **$\{2, 2, 3, 3\}$**
3. **$\{2, 2, 6\}$**
4. **$\{2, 3, 5\}$**
5. **$\{5, 5\}$**

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Python 3



```
1 n,m=map(int,input().split())
2 coins=list(map(int,input().split()))
3 dp=[0]*(n+1)
4 dp[0]=1
5 for coin in coins:
6     for i in range(coin,n+1):
7         dp[i]=dp[i]+dp[i-coin]
```

```
7         dp[i] += dp[i - coin]  
8     print(dp[n])  
9
```

Line: 9 Col: 1



Upload Code as File



Test against custom input

Run Code

Submit Code

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Test case 1

Test case 2

Test case 3

Test case 4

Test case 5

Test case 6

Compiler Message

Success